

ENPH 453 Advanced Physics Laboratory
Group 2 Letter of Intent 1

Date of submission: January 14, 2026
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Scientific Motivation & Theoretical Prediction

Muons are fundamental subatomic particles that are a part of the lepton group, possessing a heavier mass and shorter lifespan ($\tau \approx 2.2\mu s$) than the electron [1]. When high-energy charged particles from outer space (cosmic rays) interact with the atmosphere, they produce muons through pion decay [1]. These muons travel at relativistic speeds ($v \sim 0.98c$) and reach the surface of the Earth before the end of their lifespan, with a flux of $\sim 1 cm^{-2} min^{-1}$ [2]. This is because the proper time τ elapsed in the muon's rest frame is significantly less than the time T measured in the Earth frame ($T = \gamma\tau, \gamma$ being the Lorentz Factor), enabling them to survive long enough to be detected [3].

When a muon slows down inside a plastic scintillator, the deposited energy will generate a brief flash of light [4]. If it stops, it will decay into an electron (or positron) and two neutrinos, as shown below in Eq. 1, emitting a second flash of light.

$$\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e \qquad\qquad \mu^+ \rightarrow e^+ + \bar{\nu}_\mu + \nu_e \tag{1}$$

Additionally, negatively charged muons can be captured by surrounding atoms, forming a secondary decay channel and resulting in a shorter effective lifespan [3]. The stopping and decay events will appear as two sharp, coincident scintillation signals separated by a short delay [4]. By measuring the time between these events for many muons, we expect the collected data to produce an exponential decay curve, from which the mean lifetime can be extracted [2]. With a sufficiently large dataset, it may also be possible to distinguish between contributions from positively and negatively charged muons.

Apparatus & Measurement Plan

The plastic scintillator will be coupled to a photo-multiplier tube (PMT), converting the flashes of light to an electronic signal [5]. This signal will be processed by a Constant Fraction Discriminator (CFD), which suppresses background noise by rejecting pulses below a set voltage threshold [6]. The discriminator outputs will serve as the START and STOP inputs to a Time-to-Amplitude Converter (TAC), which converts the time between pulses into an analog voltage [6]. This output will be digitized by a Multi-Channel Analyzer (MCA) through a Gate and Delay Generator, with the delay adjusted so that the gate coincides with the stabilized portion of the TAC output pulse [6].

In the first week, we will verify proper functionality and calibrate each device using an oscilloscope [7]. The voltage supplied to the PMT will be varied until pulse events are observed [7]. The PMT output will then be connected to the CFD, and the event rate will be measured using a ratemeter while the discriminator threshold is increased [2]. The CFD threshold will be adjusted to a value at which the measured event rate reaches a stable plateau, indicating effective suppression of background noise [2]. Next, the range of the TAC will be set above the expected muon lifetime and connected to a Time Calibrator, which can generate stable pulses at a predefined frequency [2]. By inputting known pulse separations and measuring the corresponding TAC output voltages, a conversion factor between voltage and time will be determined [2]. The output pulse will also be examined to determine a suitable gate delay. Finally, the TAC will be connected to the MCA and we will run an initial measurement of one day using the provided LabView program. If this is successful, a longer measurement period will be performed over the subsequent 2 weeks.

References

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